

Dongsu Lee

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RESEARCH INTEREST

Reinforcement learning, multi-agent coordination, test-time computation, continual learning, robotics, vla.

RESEARCH EXPERIENCE

Jun. '26 – Aug. '26	Amazon Internship, Supply chain optimization technologies (SCOT) Working on test-time computation of safe RL with D. Richards, H. Waldon, S. Andaz	New York, NY, USA
Sep. '24 – Feb. '25	SAFE AI Lab, Carnegie Mellon University Visiting Scholar, School of Computer Science Working on communication for multi-agent coordination [P1] with Prof. D. Zhao and Y. Niu	Pittsburgh, PA, USA
Mar. '22 – Aug. '24	BMIL Lab, Soongsil University Graduate researcher Working on autonomous driving [C1] and reinforcement learning [C2, C3] with Prof. M. Kwon	Seoul, South Korea

EDUCATION

Sep. '25 – Present	University of Texas at Austin Ph.D. student, Electronics and computer engineering Advisor: Prof. Amy Zhang	Austin, TX, USA
Mar. '16 – Feb. '22	Soongsil University B.S., Bio-medical System; Big-data Computing (summa cum laude) Leave of absence for military service: Aug 2017 - May 2019 (2 years)	Seoul, South Korea

PUBLICATIONS

Conference Proceedings

C8 # Position # MARL # Test-time # Multi-agent orchestration	Beyond Training Time, Test-Time Coordination is Essential for Cooperative MARL Dongsu Lee *, Sooraj Sathish*, Joonkyung Kim, Woojun Kim, Amy Zhang <i>Advances in Neural Information Processing Systems (NeurIPS) Position Track</i> , 2026.
C7 # Test-time # Coordination # MARL # Flow matching	Test-time Multi-agent Coordination by Decomposed Value Gradient Flow Dongsu Lee , Haoran Xu, Amy Zhang <i>Advances in Neural Information Processing Systems (NeurIPS)</i> , 2026.
C6 # Memory # Robotics # Skill learning	Reconfiguring Procedural Knowledge for Compositional Robot Skill Adaptation Daehee Lee, Dongsu Lee , Sanghyun Ahn, Minjong Yoo, Honguk Woo <i>Advances in Neural Information Processing Systems (NeurIPS)</i> , 2026.
C5 # MARL # Flow matching	Multi-agent Coordination via Flow Matching Dongsu Lee , Daehee Lee, Amy Zhang <i>International Conference on Learning Representation (ICLR)</i> , 2026.
C4 # Continual learning # Robotics # Skill learning # Imitation learning	Policy Compatible Skill Incremental Learning via Lazy Learning Interface Daehee Lee, Dongsu Lee , TaeYoon Kwack, Wonje Choi, and Honguk Woo <i>Advances in Neural Information Processing Systems (NeurIPS)</i> , 2025. Spotlight .
C3 # Model-based RL # offline # Representation	Temporal Distance-aware Transition Augmentation for Offline Model-based Reinforcement Learning Dongsu Lee , Minhae Kwon <i>International Conference on Machine Learning (ICML)</i> , 2025.

C2 # MARL # Theory-of-mind
Inverse RL # Episodic reasoning

Episodic Future Thinking Mechanism for Multi-agent Reinforcement Learning

Dongsu Lee, Minhae Kwon

Advances in Neural Information Processing Systems (NeurIPS), 2024.

C1 # Autonomous driving # RL
Dataset & Benchmark

AD4RL: Autonomous Driving Benchmarks for Offline Reinforcement Learning with Value-based Dataset

Dongsu Lee, Chanin Eom, Minhae Kwon

IEEE International Conference on Robotics and Automation (ICRA), 2024.

Preprints

P3 # Zero-shot coordination # JEP
Self-play # Successor feature

Belief-conditioned Predictive Latent Embedding for Zero-shot Multi-agent Coordination

Dongsu Lee, Lucas Maes, Quentin Le Lidec, Randall Balestrieri, and Amy Zhang

NeurIPS PTA Workshop, 2026 and under review.

P2 # Offline MARL # Benchmark
Value decomposition # Analysis

A Recipe for Stable Offline Multi-agent Reinforcement Learning

Dongsu Lee, Daehee Lee, and Amy Zhang

arXiv preprint, 2026.

P1 # MARL # Communication
Representation

Unifying Agent Interaction and World Information for Multi-agent Coordination

Dongsu Lee, Daehee Lee, Yaru Niu, Honguk Woo, Amy Zhang, and Ding Zhao

NeurIPS 2025 ARLET Workshop, 2025. **Oral Presentation.**

AWARDS AND SCHOLARSHIPS

2026

Top (Gold) Reviewer

2026 International Conference on Machine Learning (ICML)

Aug. '25 – Present

UT Austin Engineering Outstanding Fellowship

Cockrell School of Engineering and The University of Texas at Austin Graduate School

Aug. '23 - Feb. '25

Hyundai Future Industrial Talent Scholarship

Hyundai Motor Chung Mong-Koo Foundation

2023

ICT Challenge IITP President's Award

Ministry of Science and ICT and Institute for Information & communication Technology Planning & evaluation

2022

Soongsil Graduate Admission Excellence Scholarship

Soongsil University

ACADEMIC SERVICES

2025 – Present

Conference Reviewer

ICLR, ICML, NeurIPS, AAAI, CoRL, RLC, ICRA, IROS, IEEE IV.

2024 – Present

Journal Reviewer

TMLR, Nature Communication, IEEE TCYB, IEEE TITS, IEEE TCE, IEEE IoTJ, IEEE TVT.